Dr Xingran “Gillian” Gao

CSci · CEnv · MIMMM

Climate, Nature & ESG Risk Analyst — Environmental Scientist

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UK Indefinite Leave to Remain — no visa sponsorship required

PROFESSIONAL PROFILE

Environmental scientist and risk analyst with a University of Cambridge PhD and Chartered Scientist status, specialising in climate, nature, ESG and water risk. Combines quantitative modelling and data science — Python, Google Earth Engine, geospatial analysis and cloud workflows — with regulatory, decision-focused consulting. Builds and owns production environmental-risk models aligned to TCFD and TNFD, and communicates technical evidence and uncertainty to senior stakeholders including the Environment Agency and Defra.

CORE EXPERTISE

Programming & data	Python (pandas, numpy), MATLAB, SQL-style querying, Google Earth Engine, BigQuery, Power BI, Excel; reproducible HPC & cloud (AWS / GCP) workflows; API integration.
Modelling & analysis	Quantitative environmental-risk modelling, geospatial analysis (QGIS, Tygron), hydrological & water-quality modelling (SOAF, UPM), climate modelling (WRF ensembles), machine learning, statistical analysis, scenario & sensitivity analysis, cost-benefit analysis.
Domain & frameworks	Climate risk (physical & transition), nature & biodiversity risk, ESG, natural capital, decarbonisation pathways; TCFD, TNFD, CSRD, Ofwat / PR24, AMMA, ISO 55001, FCERM, CRREM, EU Taxonomy.
AI & professional	Multi-agent systems (LangGraph, AG2); project management & delivery (scope, budget, QA), stakeholder & regulatory engagement, technical reporting, mentoring.

PROFESSIONAL EXPERIENCE

Risk Analyst — **Risilience** — Cambridge, UK · Jun 2024 – Present

- Design, implement and own quantitative models of nature-related, physical climate and transition risk across multiple sectors, aligned to TCFD and TNFD, with full responsibility for model development, QA and continuous improvement.
- Developed composite environmental indicators, including an Ecosystem Integrity Index, converting complex global datasets into consistent, transparent, decision-ready risk layers.
- Built and managed large-scale global water-stress datasets using Google Earth Engine and BigQuery, and developed geospatial pipelines integrating climate, hydrological and land-use data.
- Led climate-risk model development for agricultural commodity supply chains and designed scenario-based models quantifying revenue sensitivity to supply-chain and circular-economy transitions.
- Manage projects end-to-end (£3k–£50k; scope, delivery, QA), present findings to senior stakeholders, and support internal QA and the development of junior team members.

Research Associate (part-time, project-based) — **University of Cambridge** — Cambridge, UK · Jan 2026 – Present

- Lead a Tygron 3D geospatial “digital twin” of Fenland peatland, linking water-table depth to CO_2/CH_4 emissions, soil carbon and land stability under climate scenarios.
- Present modelling results and funding justifications to senior stakeholders including the Environment Agency and Defra, contributing directly to project-expansion and funding decisions.
- Established an international research collaboration with partners in the Netherlands, initiating joint publications.

Consultant, Water & Environmental Modelling — **Arup** — UK · Jun 2023 – Jun 2024

- Developed a Python-based water-quality modelling tool for Welsh Water aligned with SOAF and UPM, integrating sewer-spill time series, river-flow statistics and pollutant dynamics (DO, ammonia, BOD) for auditable regulatory decision-making.
- Co-authored an independent Asset Management Maturity Assessment (AMMA) review for Northumbrian Water, benchmarked to the Ofwat framework and ISO 55001 / SAMP, supporting its PR24 submission.

- Contributed to flood and coastal resilience Outline Business Cases applying FCERM cost-benefit and option appraisal — including a £1M scheme protecting 579 homes and 1,888 ha of land at a 31:1 benefit-cost ratio.
- Delivered climate-adaptation pathways and long-term scenario analysis for river catchments across AMP7 water-infrastructure programmes.

Engagement Manager — **Cambridge Consulting Network** Cambridge, UK · Oct 2020 – Jun 2021

- Led consulting projects for property-investment and financial-data clients, including structured social research and a cost-analysed database of 93 global stock exchanges.

EDUCATION

PhD, Environmental Science — **University of Cambridge** Oct 2018 – May 2023

Machine learning and atmospheric physics applied to urban climate (LoHCool programme); designed a 16-member WRF ensemble and processed 30+ TB of data across HPC and cloud.

MSc, Environmental Engineering — **Imperial College London** 2017 – 2018

GPA 3.6/4.0. Water demand analysis (MATLAB, Monte Carlo), wastewater and landfill design.

BSc (Hons), Environmental Science — **University of Manchester** 2015 – 2017

First Class Honours, GPA 3.8/4.0. First Class Scholarship (2015–2017); Excellent Graduate Thesis (2017).

PROFESSIONAL REGISTRATION

Chartered Scientist (CSci) — awarded 2026, IOM3 / Science Council.

Chartered Environmentalist (CEnv) — awarded 2026, IOM3 / Society for the Environment.

Chartered Engineer (CEng) — application in progress (IOM3 Technical Report Route).

MIMMM — Member, Institute of Materials, Minerals and Mining.

Member of CIWEM and CIWM.

ACHIEVEMENTS & RECOGNITION

- **2nd place**, Multi-Agent Systems for Scientific Research Hackathon, University of Cambridge (2026).
- Three Chinese invention patents (first inventor) in wastewater treatment.
- First Class Honours and First Class Scholarship; Excellent Graduate Thesis (Manchester).
- Specialist training at the European Centre for Medium-Range Weather Forecasts (ECMWF).

SELECTED PROJECTS

- **Fens Peatland Digital Twin** — lead a Tygron 3D geospatial model linking water-table depth to greenhouse-gas emissions and land stability under climate scenarios; outputs presented to the Environment Agency and Defra.
- **Global water-stress & nature-risk models (Risilience)** — built large-scale global water-stress datasets and an Ecosystem Integrity Index, turning complex data into decision-ready risk layers aligned to TCFD and TNFD.
- **Welsh Water SOAF water-quality tool** — Python tool implementing 99th-percentile and Fundamental Intermittent Standards logic for auditable, regulator-ready investment evidence.
- **PhD — aerosol impacts on urban climate (LoHCool)** — 16-member WRF ensemble across observational, satellite, reanalysis and climate-model data; quantified aerosol versus greenhouse-gas effects on urban temperature.

RESEARCH & SELECTED PRESENTATIONS

Research programmes: Low Carbon Climate-Responsive Heating and Cooling of Cities (LoHCool); Making Safer Emergency Hospitals; aerosol influence on current and projected urban temperatures. Invited talks and forums include the Cambridge Young Scholars Forum, the Green Economy Forum, and Climate Change & Carbon Net Zero (Eastern NewEnergy Accelerator Programme).